

TECHNICAL DOCUMENT

Software and Research Tools

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Tools Actually Used

Tool	Version	Role	Official source
LTspice	26.0.2	Transistor-level simulation	analog.com/ltspice
Python	Installed 3.x runtime	Verification, parsing, PDF, Word, and figure generation	python.org
Git	Installed client	Version control	git-scm.com
Icarus Verilog	12.x	SystemVerilog reference simulation	bleyer.org/icarus
GitHub	Public repository service	Public dissemination	github.com
PTM	65-nm beta model	Predictive transistor model	mec.umn.edu/ptm
ReportLab	Installed Python library	Academic PDF generation	reportlab.com
python-docx	Installed Python library	Editable Word document generation	python-docx.readthedocs.io
Pillow	Installed Python library	Research figures	python-pillow.org

Reproducibility Notes

Versions and commands are documented in the repository. Verilog source is included and can be simulated with Icarus Verilog for the reference encoder testbench.